Data Mining Assignment 2: Recommender Systems

Simple Movie Recommendation Report

GitHub repository: `[add repository link here]`

1. Introduction

The goal of this assignment is to build and compare recommender models for movie recommendation, and then generate 10 recommendations for each user in `ratings_test.csv`. I focused on the two things that the assignment asks for most clearly:

- rating prediction quality, measured with `RMSE`
- top-10 recommendation quality, measured mainly with `Precision@10` and `Recall@10`

I compared three models: a bias-based baseline, a user-based collaborative filtering model, and an SVD matrix factorization model. The final system also had to handle cold-start users, because some users in the test file do not appear in the training data.

2. Data Understanding And Preprocessing

The training data contains `97,801` ratings from `600` users on `9,680` movies. The user-item matrix is about `98.32%` sparse, so most possible user-movie pairs are missing. This is important because sparse data usually makes neighborhood methods less stable and gives more room for latent-factor methods to help.

Some other statistics are also useful:

- average ratings per user: `163.0`, median: `69.5`
- average ratings per movie: `10.1`, median: `3.0`
- share of ratings from the top `10%` most-rated movies: `59.66%`
- movies with zero training ratings: `62`

These numbers show a clear long-tail structure. A small set of movies gets many ratings, while most movies receive only a few. Because of that, popular movies are easier to recover, so popularity-heavy recommendations can look stronger than expected.

During preprocessing, I checked missing values, duplicate user-movie ratings, and the rating range. No missing values and no duplicate ratings were found. I also added simple movie statistics such as average rating and rating count, which were later useful for analysis and for the cold-start fallback.

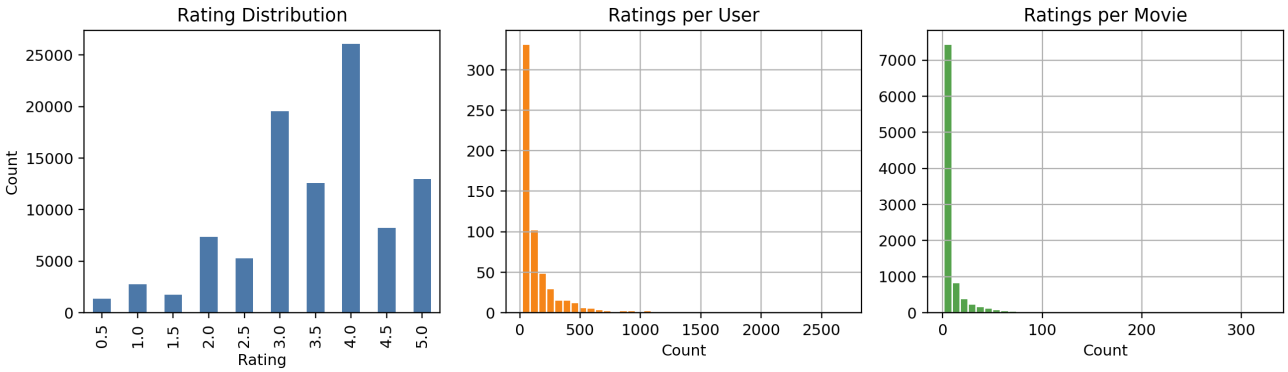

Figure 1: Basic data overview. The plots show rating concentration, uneven user activity, and a strong long-tail on the movie side.

Finally, I checked whether the users in `ratings_test.csv` were present in the training data. There were `90` warm users and `10` cold-start users, so `10%` of the final recommendations required a fallback strategy.

3. Models And Evaluation Setup

For validation, I used a temporal split. For each user, the ratings were sorted by timestamp and the last `5` interactions were held out. The remaining ratings were used for training. I chose this because the timestamps make it possible to respect interaction order instead of using a fully random split.

For ranking evaluation, only held-out ratings of `4.0` or higher were treated as relevant items. In this setup, `1,696` held-out ratings were relevant, and `533` out of `600` users had at least one relevant held-out item. This means `88.83%` of users contributed to the ranking evaluation.

The three models were:

Baseline. This model predicts ratings from the global mean together with user and item bias terms. It gives a simple reference point and helps separate real collaborative effects from global popularity and bias effects. In Surprise, this is `BaselineOnly`, which predicts from $\mu + b_u + b_i$ [3].

User-based collaborative filtering. This model estimates a user's preference from similar users. It is intuitive and easy to explain, but it depends strongly on overlap between users. In sparse data, that overlap can be weak.

SVD matrix factorization. This model learns latent user and item factors together with bias terms. It is a standard matrix factorization approach and is often more suitable for sparse recommender data because it does not rely as directly on local user-user overlap [1, 2].

I reported `RMSE`, `Precision@10`, and `Recall@10` as the main metrics because they match the assignment most directly. I kept `HR@10` and `NDCG@10` as supporting metrics only.

4. Results And Discussion

The main validation results are shown in Table 1.

Table 1: Main validation results

Model	RMSE	P@10	R@10	HR@10	NDCG@10	Time
Baseline	0.9389	0.0756	0.2255	0.5488	0.1761	0.3
User-Based CF	0.9398	0.0370	0.1137	0.3114	0.0711	1.8
SVD	0.9366	0.0638	0.1894	0.4883	0.1424	0.4

At first glance, the baseline looks surprisingly strong, especially on `Recall@10`. I think a likely explanation is the long-tail structure of the data. Since nearly `60%` of all ratings come from the top `10%` most-rated movies, popular items are easier to recover. A model that captures global movie effects can therefore perform well on ranking metrics even without much personalization.

To make this trade-off more visible, I also compared the recommendation lists themselves across `50` sampled users.

Table 2: Recommendation statistics over 50 sampled users

Model	Coverage	Avg. popularity	Max repeat
Baseline	43	129.4	46
User-Based CF	252	14.1	15
SVD	160	88.1	17

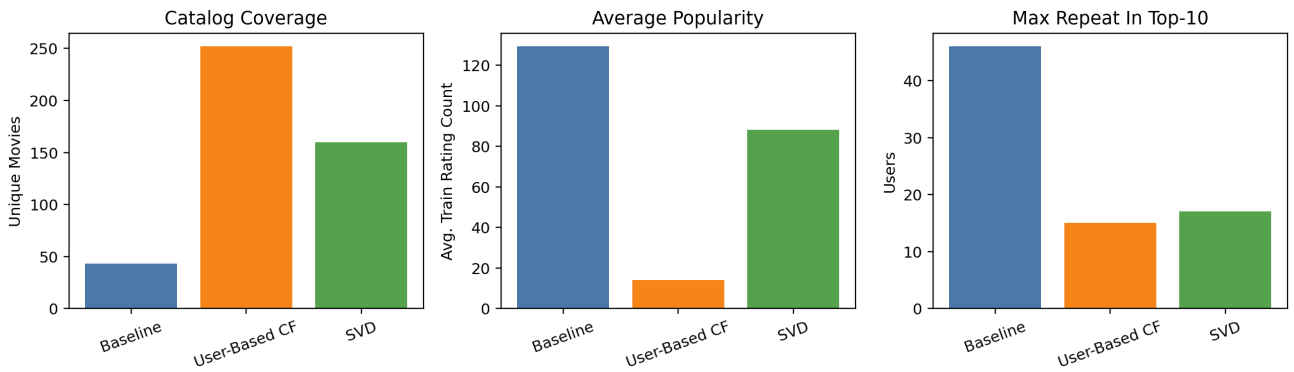

Figure 2: The baseline is strong but highly concentrated, while SVD gives a broader recommendation set without collapsing to extremely niche items.

These results help explain the model behavior more clearly. The baseline repeats the same popular movies for many users. The user-based model is much more personalized, but its accuracy is clearly weaker. SVD sits between the two and gives a more convincing compromise.

In this dataset, the user-based model suffers from the data structure itself: the matrix is about `98%` sparse and the median movie has only `3` ratings, so user-user overlap is often too limited for stable neighborhoods. SVD gives the best `RMSE`, clearly better ranking performance than the user-based model, and much broader coverage than the baseline. For that reason, I selected SVD as the final personalized model.

I also tested a small hyperparameter search for SVD. However, it was optimized for `RMSE` on random folds and did not improve `Precision@10` or `Recall@10` on the temporal validation setup, so I did not use it as the final model.

5. Final Recommendation Strategy

For the final recommendation step, I retrained SVD on the full training set and generated the top `10` unseen movies for each warm user.

The final pipeline handled two cases:

- **Warm users:** score all unseen movies with SVD and return the top `10`
- **Cold-start users:** return a popularity-based fallback list

The fallback was needed for `10` out of `100` test users. For those users, collaborative filtering cannot personalize recommendations because there is no interaction history. Instead of using a plain average rating, I used a weighted rating formula so that movies with only a few ratings would not look artificially strong.

This is not a perfect solution to the cold-start problem, but it is simple, transparent, and reasonable for the scope of the assignment.

6. Conclusion

This dataset is challenging because it is highly sparse and strongly long-tail. Those properties make popularity-based recommendations look stronger than expected and make user-based neighborhoods less stable.

Among the tested models, SVD was the most suitable final personalized model for this assignment. It had the strongest `RMSE`, better ranking results than user-based CF, and much broader recommendation coverage than the baseline. The final system therefore uses SVD for warm users and a weighted popularity fallback for cold-start users.

I do not claim that this fully solves recommendation quality or cold-start. However, I think it is a logical and well-motivated solution for the scope of the assignment.

References

[1] Y. Koren, R. Bell, and C. Volinsky. Matrix factorization techniques for recommender systems. *Computer*, 42:30–37, 2009. https://www.chrisvolinsky.com/publications/17545-matrix-factorization-techniques-for-recommender-systems

[2] Surprise documentation. Matrix factorization-based algorithms. https://surprise.readthedocs.io/en/stable/matrix_factorization.html

[3] Surprise documentation. Basic algorithms. https://surprise.readthedocs.io/en/stable/basic_algorithms.html